

From Transfer of Possession to Modification: The Grammaticalization of Chinese DE^{*}

Mingyang Bian

Qualifying Paper 2

Committee Members: Paloma Jeretič, Don Ringe

Abstract

This paper investigates the diachronic development of the Chinese character DE, which surfaces in Modern Mandarin with multiple seemingly unrelated functions: a transfer-of-possession verb, a possibility modal, a necessity modal, and a post-verbal structural linker in the V-DE construction. Focusing on the V-DE construction, I argue that synchronic analyses are negligent of whether a unified analysis of DE is expected and propose a two-pathway architecture of grammaticalization, supported by converging interpretational, chronological, and distributional evidence from historical corpora spanning Late Archaic Chinese to Pre-modern Mandarin. Along the first pathway, the V-DE-NP construction emerged in 9–10th *c.* CE when the transfer-of-possession verb DE entered serial verb constructions; V-DE-CP arose concurrently via NP/CP argument-type flexibility, and V-DE-AP developed by 13–14th *c.* CE through structural reanalysis of V-DE-CP with null embedded subject and adjectival predication. Along the second pathway, the modalized V-DE-V construction emerged centuries later, in 19th *c.* CE, derived from the possibility modal DE rather than from the lexical verb DE. I provide formal denotations for the V-DE-NP/CP/AP branch building on the semantics proposed for lexical verb DE in Bian & Jeretič 2026, and speculate on the development of the V-DE-V branch by comparing two hypotheses and discussing potential solutions to the related theoretical puzzles. The study illustrates how diachronic evidence can inform synchronic compositional analysis by adopting a diachronic formal semantics methodology where compositional choices are consequences of independently motivated structural changes.

^{*}I would like to extend my sincere gratitude to Paloma Jeretič and Don Ringe for many insightful comments, suggestions, and support throughout various stages of this project. I am also grateful to my audience in Penn Meaning Lab for their helpful feedback and questions. All errors are my own.

1 Introduction

The Chinese character 得 (Modern Mandarin *dé, de, děi*; Karlgren–Li reconstruction of Middle Chinese /*tək/) presents a compelling case study in both the syntax-semantics interface and historical linguistics, illustrating how a single logographic unit can diverge into distinct lexical and functional items with different sound forms over time. In Modern Mandarin, this character exhibits three distinct sound forms that correlate with specific syntactic-semantic functions. It appears as *dé* (/tɤ35/) when functioning as a lexical, transfer-of-possession verb meaning ‘to get; to obtain’ or a possibility modal, as *de* (/tɤ3/) with the neutral tone when serving as a structural linker between a verb and another constituent, and as *děi* (/tɛi214/) when used as a necessity modal.

In the current paper, we focus on the post-verbal structural linker use of 得, which is commonly called ‘the v-DE construction’ in the literature; 得 is often treated as a marker of modification. This type of use shows interesting distributional and interpretational patterns depending on the category of the complement it takes. When the complement following v-DE is a nominal, the verb and the post-verbal linker 得 are highly lexicalized and no longer analyzable for speakers who are not familiar with Classical Chinese, and 得 largely retains its transfer-of-possession verb meaning ‘to get; to obtain’:

- (1) Zhangsan huò-de/qu-de/ying-de le yì cháng shènglì.
Zhangsan acquire-DE/fetch-DE/win-DE PFV one CL victory
‘Zhangsan has acquired/fetched/won a victory.’

When the complement following v-DE is a verb phrase, the verb phrase is interpreted as the potential result of the verb preceding 得. Interestingly, when the v-DE-v construction is negated, 得 is simply replaced by negation:

- (2) a. Zhangsan chī de wán zhè wǎn fàn.
Zhangsan eat DE finish this CL rice
‘Zhangsan can eat this whole bowl of rice.’
b. Zhangsan chī bù wán zhè wǎn fàn.
Zhangsan eat NEG finish this CL rice
‘Zhangsan cannot eat this whole bowl of rice.’

When the complement following v-DE is a full clause, the clause is interpreted as the actual result of the preceding verb:

- (3) tā dà de wǒ tóu téng.
he hit DE I head hurt
‘He hit me; as a result, I got a headache.’ (W. Li 2024)

Finally, v-DE can take adjectival modifiers as its complement. This subtype of v-DE construction has received particular attention in the literature because it shows different structures and interpretations with the same surface patterns. For example, following Huang 1988 and Y.-H. A. Li 1990, there is a distinction between descriptive complement constructions and resultative complement construction.

- (4) a. wǒ pǎo de hěn kuài.
I run DE very fast
‘I run very fast.’

- b. tamen tiao de hen lei.
 they jump DE very tired
 ‘They jumped till they got very tired.’ (Huang 1988)

(4-a) is an example of descriptive complement construction, where the adjective serves as a degree or manner modifier of the verb preceding DE; on the other hand, (4-b) is an example of resultative complement construction, where the adjective describes the state of the subject as the result of performing the verb preceding DE. Besides V-DE-AP, V-DE-CP can also form resultative complement construction, where the second predicate has its own subject:

- (5) tamen ku de shoupa dou shi le.
 they cry DE handkerchief also wet ASP
 ‘They cried so much that even the handkerchief got wet.’ (Huang 1988)

Previous studies have proposed various classifications and analyses for the V-DE construction from a synchronic perspective on Modern Mandarin (see Huang 1988, Xie 2012, C. Li 2015, W. Li 2024 for direct engagement with the construction) and Cantonese (Lam 2023); to my knowledge, no previous proposal has considered the implications of the diachrony of DE on analyzing the syntax and semantics of the V-DE construction, which has resulted in some theoretical gaps that the current paper aims to fill: first, whether the numerous seemingly unrelated constructions such as transfer of possession, modal constructions, and resultative/descriptive complementation are derived from the same lexical source, and if so, how; second, how different subtypes of the V-DE construction are derived and connected with each other, and whether a unified analysis of DE is expected.

For example, Huang 1988 examines the V-DE construction from a syntactic point of view and argues for an analysis where the verb preceding DE is the main predicate of the sentence, and the complement of V-DE is an adjunct or secondary predicate; DE itself is treated only peripherally as a clausal complementizer, an AP marker, or a verbal suffix/clitic, with no commitment to a semantics of DE. For Huang, the resultative/descriptive distinction shown in (4) follows from the synchronic syntax, and therefore they are completely different syntactic structures that happen to both have DE as a structural linker (or modification marker) between the main predicate and the adjunct/secondary predicate. The study does not discuss the modalized V-DE-V construction, and it is not clear whether the proposal can be extended to this subtype. C. Li 2015 proposes a two-tier classification orthogonal to the Huang tradition and treats DE as a verb suffix which is purely functional and lacks a concrete semantics. In this study, there is no diachronic argumentation beyond the claim that the verb-to-affix grammaticalization of DE ‘is consistent with a well-established path’, citing Hopper & Traugott 2003; the modalized V-DE-V subtype is not discussed as well. On the other hand, Xie 2012 focuses exclusively on the modalized V-DE-V subtype, proposing the formal semantics of modal DE in V-DE constructions. However, the study offers no discussions on other subtypes of the V-DE construction or on modal constructions with DE in general. Lam 2023 proposes a unified analysis for the V-DE construction in Cantonese (DE is realized as *dāk*) where DE heads a ModP specified with modality features, which in turn takes as complement an AspP with a [$\pm$ realize] feature that determines whether the interpretation involves modality (as in the V-DE-V subtype) or not (as in the descriptive/resultative V-DE-AP/CP constructions). However, Lam 2023 does not offer any speculations on whether or how DE in the V-DE construction is derived from the transfer-of-possession verb DE, despite adopting the label ACQ(UIRE) for DE throughout the study.

The current paper claims that the seemingly unrelated functions of DE are all derived from one single lexical source of DE as a transfer-of-possession verb and follow a two-pathway architecture of grammatical-

ization for the V-DE construction based on chronological, interpretational, and distributional evidence. Section 2 of the paper presents the results from our historical corpora containing data points with DE in various periods of history, based on which I propose a chronology of the development of DE and spell out the two-pathway architecture for its grammaticalization. Section 3 draws on Shi 2023 and Shi 2025 and places the grammaticalization process of DE in the broader picture of the development of verbal phrases in Chinese, arguing that the loss of predicate and clause coordination and the emergence of verb serialization and resultative construction provided the structural basis of the V-DE construction. In Section 4, I offer a formal analysis of three subtypes of the V-DE construction, namely V-DE-NP, V-DE-CP, and V-DE-AP, which are argued to be derived directly from the transfer-of-possession verb DE and constitute one of the two branches of the two-pathway architecture. In Section 5, I argue that the one remaining subtype of the V-DE constructions, namely V-DE-V, is derived from the possibility modal DE (which is in turn derived from the transfer-of-possession verb DE *à la* Bian & Jeretič 2026) and constitutes the other branch of the two-pathway architecture; I will then speculate on the development process with this branch by comparing two hypotheses. Section 6 concludes the paper.

2 Historical Corpora

This section illustrates the empirical motivations for going beyond the previously proposed synchronic analyses of the V-DE construction.

2.1 Methodology

We track the diachronic development of DE by collecting pre-modern Chinese use of DE from historical sources, drawing from texts in different periods of the development of the Chinese language. We establish and classify our corpora according to the era of each text source being examined, focusing on written Colloquial Chinese rather than Literary Chinese since the former more accurately reflects the dynamics of language change.¹ The text sources we considered cover several eras between Late Archaic Chinese (5th to 2nd *c.* BCE) and Pre-modern Mandarin (19th to 20th *c.* CE), with particular attention to the following critical periods:

1. Late Archaic Chinese, when classic, philosophical texts flourished in the Warring States period, providing more reliable text sources compared to oracle bone and bronze scripts from preceding periods of Early Archaic Chinese (10th to 6th *c.* BCE).
2. The early stage of Late Middle Chinese (circa 9th to 13th *c.* CE), when Colloquial Chinese began to appear in written forms substantially in Buddhist vernacular tales (*Bianwen*) after the divergence of Literary and Colloquial Chinese towards the end of the Han Dynasty (circa 3rd *c.* CE).
3. The transition from Old Mandarin (13th to 14th *c.* CE) to Middle Mandarin (14th to 19th *c.* CE), when vernacular literature fully matured and provides abundant evidence for diachronic analysis of these eras.

¹See C. W.-C. Li 2016 for an introduction of the history of diglossia in Classical Chinese. To put it briefly, towards 1st-3rd *c.* CE, Literary and Colloquial Chinese began showing significant divergence. However, Colloquial Chinese was not documented in written forms substantially until the growing popularity of Buddhist transformation texts (vernacular tales) in late Tang Dynasty (circa 9-10th *c.* CE). The diglossia ended in 20th *c.* CE, when Literary Chinese was completely replaced by Colloquial Chinese for written texts following the New Culture Movement.

We collect texts from historical sources digitalized by Chinese Text Project and Wikisource. In our historical corpora, we annotate each occurrence of DE for its use (lexical verb, modal, post-verbal linker in V-DE). We have established that the interpretational and distributional differences shown by the subtypes of the V-DE construction are primarily determined by the category of the complement; therefore, for the post-verbal linker uses specifically, we further annotate the category of the complement of V-DE: NP, CP, V, or AP. For example:

- (6) ta you yi ge nv'er, jin'nian qi sui, sheng de kexi, zhang de ke'ai.
 he have one CL daughter this.year seven year.old born DE delightful grow DE lovely
 'He has a seven-year-old daughter who is delightful and lovely.'
 (*Dou'e Yuan*, mid 13th c. CE)

In this data point, we have two occurrences of DE, and both are post-verbal and are followed by an adjectival modifier. Therefore, we annotate both uses of DE as post-verbal linker with AP as the complement category.

For each period considered, we also calculate the frequency of each complement category if the corpus size of that period is large enough for meaningful comparison and document the earliest presence of each complement type and notable distributional patterns (e.g., co-occurring particles, syntactic environments, verbs preceding DE, etc.).

2.2 Results and sample data points

For 5th-2nd c. BCE (Late Archaic Chinese), we primarily considered sections from the Confucianist classics *Lunyu* (The Analects) and *Liji* (Book of Rites). In 63704 characters, 95 occurrences of DE were found, with 48 lexical verb uses and 47 modal uses; there is no instances of the V-DE construction.²

- (7) a. qiu ren er de ren, you he yuan?
 seek benevolence then DE benevolence again what complaint
 'Those who seek benevolence and attain it — what complaint could they have?' (*Lunyu*)
 b. bu de qi jiang, bu shi.
 NEG DE PRON sauce NEG eat
 '(Talking about food) If one does not get the proper sauce, they do not eat (the food).'

(*Lunyu*)

The examples above illustrate the lexical verb uses of DE in this period, which functions as a transitive verb of transfer of possession.

- (8) a. de jian junzi zhe, si ke yi.
 DE see gentleman NOM then sufficient P
 'To get to meet a true gentleman - that alone is enough.'

(*Lunyu*)

²We did not consider texts from Early Archaic Chinese (10th to 6th c. BCE), but Meisterernst 2019 does report that DE is only attested as a lexical verb with the basic meanings 'to get; to obtain,' etc. in Early Archaic Chinese based on evidence from *Shijing* (Classic of Poetry). Following the hypothesis in Bian & Jeretič 2026, light verb modals are formed from type-flexible existential semantics with an underspecified relation encoding possession or modality. This hypothesis captures the fact that there exists a period of time when DE only has a lexical verb use, and the possibility modal use is later derived from the lexical verb use.

- b. kongzi xia, yu yu zhi yan. qu er bi zhi, bu de yu zhi yan.
 Confucius get.off want with PRON talk leave to avoid PRON NEG DE with PRON talk
 ‘Confucius got off the carriage and wanted to talk with Jieyu, but Jieyu went away to avoid
 Confucius, so Confucius wasn’t able to talk to him.’ (Lunyu)

The two examples above illustrate the circumstantial possibility uses of DE. In both cases, the possibility is about the circumstances aligning to allow the event to occur (as opposed to rules, inherent ability, etc.).

- (9) a. she zhong ze de wei zhuhou.
 shoot hit then DE become lord
 (Talking about hunting with bows and arrows) ‘If one shoots and hits, he can become a lord.’ (The Book of Rites)
- b. tugong bu xing, dafu bu de zao chema.
 civil.engineering NEG flourish official NEG DE build chariot
 (Talking about times of natural disaster) ‘Civil engineering should be halted and officials are not allowed to build chariots.’ (The Book of Rites)

These two examples illustrate the deontic possibility uses of DE. In both examples, the possibility is about established rules (rules for becoming a lord and rules during a natural disaster).

The pattern shown by the 5th-2nd c. BCE corpus reveals an important chronological sequence: possibility modal DE precedes the emergence of the V-DE construction. In fact, the first instances of the V-DE construction appeared centuries later than the possibility modal uses; we illustrate in Section 3 that this is consistent with the general diachronic development of verbal phrases in Classical Chinese. This implies that by the time the V-DE construction came into existence, there were at least two lexical entries of DE: one is the transfer-of-possession verb, and the other is the possibility modal. This might be related to one crucial distinction we have seen with the subtypes of the V-DE construction: V-DE-V involves modality, but V-DE-NP/CP/AP does not.³ We will argue in the later sections that DE in V-DE-V is derived from modal DE, while DE in V-DE-NP/CP/AP is derived directly from lexical verb DE.

We observe the first instances of the V-DE construction in Buddhist transformation texts in 9-10th c. CE (compiled into *Dunhuang Bianwen Ji*, a collection of Dunhuang Buddhist transformation texts). In this period, the available complement categories of V-DE are restricted to NP and CP. We considered 33639 characters, and DE occurred 109 times. 39 of the occurrences are post-verbal DE, with 21 taking NP complements and 18 taking CP complements. Some sample data points are listed below:

- (10) a. zhi zhi tian ming, zao de yi si.
 straight arrive day bright build DE one temple
 ‘At sunrise, a temple was built.’ (Dunhuang Bianwen Ji)

³One might object that in Modern Mandarin, V-DE-AP can have a modalized reading as well. For example:

- (i) zhe haizi zhang de gao.
 this child grow DE tall
 Attested meaning 1: ‘This child is tall (now).’
 Attested meaning 2: ‘This child can grow tall (in the future).’

I argue that the ambiguity is due to the fact that Mandarin allows bare adjectives for non-verbal predication. Attested meaning 1 results from a V-DE-AP construction, but attested meaning 2 results from a V-DE-V construction. This sort of ambiguity partially motivates the unified treatment of modal and non-modal post-verbal DE in Cantonese in Lam 2023.

- b. dan de yi kaolao mantou...
pick.up DE one basket steamed.bun
'Picking up a basket of steamed buns, ...' (Dunhuang Bianwen Ji)
- c. dao de nan an...
arrive DE south bank
'Arriving at the south bank, ...' (Dunhuang Bianwen Ji)
- d. chu chu souxun, ruo ye zhuo de shi seng, su ling jiang lai jian wo.
place place search if also capture DE master monk quick bring P come see I.SG
'Search everywhere, and if you manage to catch the master monk, bring him to me at once.'
(Dunhuang Bianwen Ji)
- e. bian guo qi jiang, qu de xian yao, jin shang huangdi.
then pass PRON river fetch DE celestial medicine enter up emperor
'He crossed the river, acquired the elixir of immortality, and presented it to the emperor.'
(Dunhuang Bianwen Ji)

In the examples above, V-DE takes NP complements. It is worth noting that in some cases, the transfer-of-possession meaning of DE is more transparent: for example, with *qu-de* 'to fetch and obtain', DE clearly contributes the meaning of 'to obtain'; but with *dao-de* 'to arrive', the V-DE construction is purely locative, and the semantics of DE is more opaque. We take this as evidence that DE undergoes semantic bleaching with the grammaticalization process in this period, which motivates part of our formal semantic analysis in Section 4.

When taking a full clause as complement, the V-DE-CP construction in this period is very similar to that in Modern Mandarin: the complement clause encodes the actual result of the verb preceding DE.

- (11) mi de Xiang wang pao guo wei.
bewitch DE Xiang king abandon state position
'She bewitched King Xiang so that he cast aside his state position.'
(Dunhuang Bianwen Ji)

In such cases, DE is not interpreted as 'to obtain some entity'; instead, its interpretation is more aligned with 'to obtain some situation (as the result of the preceding verb)'. We will see in Section 4 that this observation motivates the two (minimally) varied lexical entries of post-verbal DE.

In 13-14th c. CE, another genre of vernacular literature drove a surge of written Colloquial Chinese texts: opera in the Yuan Dynasty (commonly called *Zaju*). Our corpus for this period of time consists of scripts of *Dou'e Yuan* (*The Injustice to Dou'e*, mid-13th c. CE) and *Xi Xiang Ji* (*Romance of the Western Chamber*, early 14th c. CE), two of the most influential *Zaju* works. We found 50 instances of post-verbal linker DE, and most notably, adjectives became available as the complement of V-DE. Out of the 50 instances, 16 take NP complements; 27 take CP complements; and 7 take AP complements. We highlight a few examples with AP complements:

- (12) a. ta you yi ge nv'er, jin'nian qi sui, sheng de kexi, zhang de ke'ai.
he have one CL daughter this.year seven year.old born DE delightful grow DE lovely
'He has a seven-year-old daughter who is delightful and lovely.'
(Dou'e Yuan)
- b. bu si zhe xiao nizi sheng de shifen beilai.
NEG resemble this little girl born DE very cheeky

‘They are not like this little girl who is very cheeky in nature.’

(*Dou’e Yuan*)

- c. bi you sheng de hao de nv’re.
must have born DE good REL woman
‘There must be women who are beautiful.’

(*Bao Daizhi Zhi Zhan Luqilang*)

One point to note about the chronological sequence of V-DE-NP/CP and V-DE-AP: V-DE-AP became available after V-DE-NP/CP, and the chronological gap is relatively short. We will argue in Section 4 that V-DE-AP diverged into two types corresponding to the descriptive complement construction and the resultative complement construction as defined in Huang 1988 in 14th c. CE.

From 15th c. CE onward, novels written in Colloquial Chinese became increasingly popular and remained the predominant literary form. The available categories of complement to V-DE were stable between 15-18th c. CE. Our corpus for 15th c. CE consists of chapters from *Shuibu Zhuan* (*Water Margin*): out of 59 instances of post-verbal DE, 20 take NP complements, 35 take CP complements, 4 take AP complements. The corpus for 16th c. CE consists of chapters from *Xiyou Ji* (*Journey to the West*): 88 instances of post-verbal DE were identified, with 34 taking NP complements, 49 taking CP complements, 5 taking AP complements. The corpus for 17th c. CE consists of sections from *Xing Shi Heng Yan* (*Stories to Awaken the World*): 14 cases of V-DE-NP; 19 cases of V-DE-CP; 9 cases of V-DE-AP. Finally, the 18th c. CE corpus consists of chapters from *Hong Lou Meng* (*Dream of the Red Chamber*), where we found 25 V-DE-NP constructions, 24 V-DE-CP constructions, and 10 V-DE-AP constructions. Some sample data points are listed below illustrating the V-DE construction with different categories of complement in these periods:

- (13) a. shi ri qing ming de hao.
this day sunny bright DE good
‘On this day, it was clear and bright to a great extent.’
b. tiao chu yi-zhi diao jing bai e dachong, jing de xiaguan hunpo
jump out one-CL hanging eye white forehead tiger frighten DE lower.official soul
dou mei le.
all NEG ASP
‘A tiger with hanging eyes and white forehead jumped out, frightening me so much that I lost my wits.’
(*Water Margin*, circa 15th c. CE)
- (14) a. meihouwang ting de ci yan, man xin huanxi.
monkey.king listen DE this word full heart happiness
‘The Monkey King heard these words, and his heart filled with joy.’
b. wo bi na ceng bing hou dong, wen de jin li.
I.SG compared.to that layer ice thick frozen steady DE to.a.great.extent SFP
‘I am comparable to that layer of thick ice, solid and unyielding to a great extent.’
(*Journey to the West*, 1520-1580 CE)
- (15) a. ben xian jiazhuang dou yi bei de shiquan.
this magistrate dowry all already prepare DE complete
‘This magistrate has already prepared the dowry completely.’
b. dao de shu bian kan shi, zhi ku ye wei.
arrive DE tree side look time branch withered leave shriveled

‘When (they) arrived at the tree to look, the branches were withered, and the leaves were shriveled.’
(*Stories to Awaken the World*, 1620-1640 CE)

- (16) a. fang tan de san wu ju hua, hu jiaren fei bao.
just talk DE three five sentence word suddenly servant fly report
‘They had only exchanged a few words when a servant suddenly rushed in with urgent news.’
b. jian ta sheng de congming junxiu, ye yu shi ta shi ji ge zi.
see she born DE bright beautiful also want make she recognize several CL character
‘Seeing that she was bright and beautiful, he wished for her to learn a few characters.’
(*Dream of the Red Chamber*, 1780-1792 CE)

The available complement category of V-DE further expanded to include V no later than 19th c. CE. We identified the first instances of V-DE-V in *Laocan You Ji* (*Journey of Laocan*, late 19th c. CE), which forms our corpus for 19th c. CE. The language used in this novel is quite close to Modern Mandarin, and post-verbal DE is particularly frequent: out of the 94 cases of post-verbal DE we annotated, 21 take NP complements; 36 take CP complements; 28 take AP complements; and 9 take V complements. In the V-DE-V construction, the second verb expresses the potential result of the first verb:

- (17) a. jizhi li da chuan shi yu zhang shi, lian chuan shang ren shuohua dou
by.the.time from large ship ten plus zhang time even ship on people talk all
ting de jian le.
listen DE see PFV
‘When they had come within about ten-odd zhang of the large ship, they could even hear the people on the ship talking.’
b. wo qu ti shaonainai shefa, zuo de cheng ye bie huanxi, zuo
I go on.behalf.of young.mistress find.a.way do DE complete also NEG.IMP rejoice do
bu cheng ye bie manyuan.
NEG complete also NEG.IMP complain
‘I’ll go and try to find a way for the young mistress. If I succeed, don’t be overjoyed; if I don’t succeed, don’t complain either.’
(*Journey of Laocan*, late CE 19th c.)

One point to note about the chronological sequence of V-DE-NP/CP/AP and V-DE-V: the latter appeared significantly later than the former; the chronological gap is several centuries wider than the gap between V-DE-NP/CP and V-DE-AP. This observation renders derivational connections between V-DE-V and other subtypes of V-DE construction implausible. Further notice that in (17-b), the V-DE-V construction can be negated by replacing DE with negation; this is consistent with the pattern we have seen with Modern Mandarin in (2-b). In contrast, the other subtypes of the V-DE, V-DE-NP/CP/AP, cannot be negated.

2.3 The two-pathway architecture

The corpus results presented in the previous subsection have several important implications: first, the V-DE construction came into existence when there were already at least two lexical entries of DE, which are the transfer-of-possession verb DE and the possibility modal DE; second, V-DE-NP and V-DE-CP are first observed in the same period of time, and both have the resultative/causal, non-modalized reading with minimal variations in terms of the interpretational property; third, V-DE-AP appeared shortly after V-DE-NP/CP and also have non-modalized reading; fourth, V-DE-V appeared much later, with distinct

- b. xue er shi xi zhi.
learn and often review PRON
'Learn and often review it.'
(*Lunyu, Xue Er*, 480-350 BCE)
- c. bu yi er fu qie gui
NEG righteous and wealthy and honored
'to be wealthy and honored in an unrighteous way'
(*Lunyu, Shu Er*, 480-350 BCE)

The requirement of coordination particles in Archaic Chinese prevents DE from being adjacent to another verb, eliminating the possibility of the V-DE construction, which is consistent with our 5th-3rd c. BCE corpus results. For example, the V-DE-NP construction is expressed by V-NP-CONJ-DE-NP, i.e., the object is spelled out twice with the two verb phrases coordinated with an overt marker:

- (19) a. qiu ren er de ren, you he yuan.
seek benevolence and DE benevolence again what complaint
'They sought benevolence and obtained it; what complaint do they have?'
(*Lunyu, Shu Er*, 480-350 BCE)
- b. min bu qiu qi suo yu er de zhi, wei zhi xin.
people NEG seek PRON NOM want and DE PRON say PRON confidence
'When the people get what they desire without seeking for it, we have the condition of confidence.'
(*Liji, Jing Jie*, 475-221 BCE)

The requirement of coordination particles gradually loosened (although the motivation is unclear), becoming optional around the first century BCE; we therefore observe instances where two verb phrases appear adjacent to each other without overt coordination:

- (20) gong zheng bai zhi.
attack Zheng defeat PRON
'(They) attacked the Zheng country and defeated it.'
(*Shiji, Zhao Shi Jia*, 100 BCE)
- (21) ji Liyou jun po zhi.
strick Liyou troop break PRON
'(They) attacked the troops of Li You and overcame them.'
(*Shiji, Cao Xiang Guo Lie Zhuan*, 100 BCE)

The development of verbal phrases stands out even more when we compare historical texts from different periods of time that describe the same historical event. In the following pairs of examples, *Zuo Zhuan*, which was written in Archaic Chinese in 550-400 BCE, describes a series of actions/events with two separate clauses, while *Shiji*, which was written in Middle Chinese in 100 BCE, adopts a serial verb construction that expresses the same series of actions without overt coordination or separation of clauses.

- (22) a. sui xi yu, mie zhi.
then raid Yu destroy PRON
'Then, (they) raided the Yu country and destroyed it.'
(*Zuo Zhuan*, 550-400 BCE)
- b. xi mie yu.
raid destroy Yu
'(They) raided and destroyed the Yu country.'
(*Shiji, Jin Shi Jia*, 100 BCE)

- (23) a. she Gong wang, zhong mu.
 shoot Gong king hit eye
 ‘(He) shot King Gong and hit his eye.’ (Zuo Zhuan, 550-400 BCE)
- b. she zhong Gong wang mu.
 shoot hit Gong king eye
 ‘(He) shot and hit King Gong’s eye.’ (Shiji, Jin Shi Jia, 100 BCE)

With the decline of overt coordination of verbal phrases, verbs frequently occur adjacent to each other, resulting in the growing popularity of serial verb constructions. Shi 2023 reports a corpus analysis on the frequency of the connective *ér* in different stages of the development of Chinese: in works of Archaic Chinese, *Lunyu* (480-350 BCE) has 160 tokens of *ér* every 10,000 characters; *Mengzi* (circa 250 BCE) has 171 tokens. In contrast, in works in Late Middle Chinese, *Liu Zu Tan Jing* (circa 700 CE) have 9 tokens of *ér* every 10,000 characters; *Shen Hui Yu Lu* (circa 750 CE) have 16 tokens. This suggests that by 8th c. CE, verb coordination with overt connectives had become increasingly unpopular; it was later completely abandoned around 10th c. CE (Shi 2023). This time point is particularly interesting because we observe the earliest instances of the V-DE construction with NP and CP complements exactly in Buddhist transformation texts from 9-10th c. CE. One natural assumption is that the transfer-of-possession verb DE enters serial verb constructions (V₁-V₂-object) as V₂ with other verbs as V₁, forming the surface V-DE construction, and the lexical semantics of DE is increasingly bleached over time, to a point where it behaves like a clitic to the preceding verb. We build on this theoretical assumption in Section 4, aiming to provide a proposal for the semantic bleaching process for lexical verb DE as V₂ in serial verb constructions.

On the other branch of our two-pathway architecture, we claim that the V-DE-V subtype of the V-DE construction is derived from the possibility modal DE, which does not fit with the process we sketched in the last paragraph. As a modal auxiliary, possibility modal DE occurs pre-verbally (and adjacent to verbs) since its emergence in Archaic Chinese, as expected:

- (24) a. de jian junzi zhe, si ke yi.
 DE see gentleman NOM then sufficient P
 ‘To get to meet a true gentleman - that alone is enough.’ (*Lunyu*)
- b. kongzi xia, yu yu zhi yan. qu er bi zhi, bu de yu zhi yan.
 Confucius get.off want with PRON talk leave to avoid PRON NEG DE with PRON talk
 ‘Confucius got off the carriage and wanted to talk with Jieyu, but Jieyu went away to avoid Confucius, so Confucius wasn’t able to talk to him.’ (*Lunyu*)

It is then puzzling how modal DE enters the V-DE-V construction. We can start approaching this puzzle by examining what is dubbed ‘the negative potential form’ in Shi 2023.

There is a large body of literature on the syntax and semantics of serial verb constructions (SVC) in Modern Mandarin, and the canonical assumption is that one important subtype of SVC is resultative constructions where V₂ strictly encodes the result of V₁ (see Tay 2024, the seminal work on Mandarin resultative expressions). Some early examples of the resultative construction with *cheng* ‘to complete’ as the result verb:

- (25) a. qiu zhi bi zuo shu, shu cheng.
 request paper pen write letter letter complete
 ‘(He) requested paper and pens to write a letter and the letter was complete.’

(*Shi Shuo Xin Yu*, 450 CE)

- b. He Pingshu zhu Laozi shi cheng.
He Pingshu annotate Laozi just complete
'He Pingshu just finished annotating Laozi.'
(*Shi Shuo Xin Yu*, 450 CE)
- c. Sun Xinggong zuo Tiantai fu cheng.
Sun Xinggong write Tiantai prose complete
'Sun Xinggong finished writing the prose of Tiantai.'
(*Shi Shuo Xin Yu*, 450 CE)
- d. lei cheng fang-zhang.
build complete Buddhist.abbey
'(They) finished building a Buddhist abbey.'
(*Dunhuang Bianwen Ji*, *Wei Mo Jie*, 9-10th c. CE)

Shi 2023 proposes that the possibility modal DE enters resultative constructions exclusively in its negative form *bu-de* (NEG-DE) as early as 13th c. CE, expressing the result of the preceding action.

- (26) a. jin zheng yao ren nendi lihui, bu de.
now just request people that understand NEG DE
'Now you requested that other people understand in that way, but you could not do so.'
- b. zhongshen gaikou bu de.
lifetime change-mouth NEG DE
'You can never take back your promise for a lifetime.'
- c. shi duo ji bu de.
thing most memorize NEG DE
'Most things cannot be memorized.'
- d. ta ji du bu de shu.
3SG already read NEG DE book
'He already cannot read books.'
(*Zhu Zi Yu Lei*, 1200 CE)

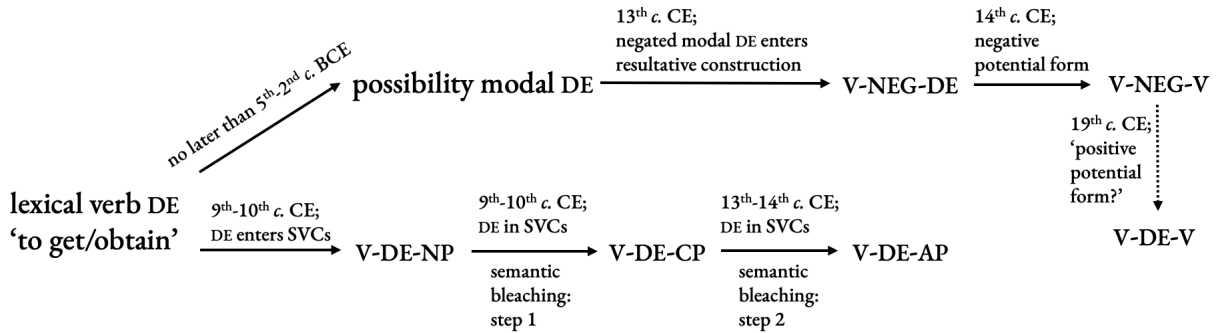
The four example sentences above represent a scale of degree of independence of the resultative *bu-de* and the preceding clause. In (26-d), the main verb *du* 'to read' is highly fused with the negated modal DE, forming a single constituent takes the object *shu* 'book'. The high frequency of this type of construction (V-NEG-DE-OBJECT) lead to the removal of the lexical constraints on DE, establishing a new grammatical device by 14th century CE dubbed by Shi 2023:129-130 as the negative potential form (V-NEG-V-OBJECT):

- (27) a. ta ye zi e ta bu zhu.
he also self contain 3SG NEG stop
'He also could not contain it.'
(*Zhu Zi Yu Lei*, 1200 CE)
- b. ni yan bu xia xiangsi zhe kou xian.
you swallow NEG down lovesick DEM CL sputum
'You cannot swallow this lovesick sputum.'
(*Yuan Kan Za Ju*, 1300 CE)
- c. suo bu zhu xin-yuan-yi-ma.
lock NEG stay fanciful.imagination
'His fanciful thinking cannot be stopped.'
(*Romance of the West Chamber*, 1300 CE)

The proposal in Shi 2023 has several potential problems. First, it is not clear in the first place why only the negated possibility modal DE can enter resultative constructions while the positive form cannot. Second,

there is no explanation on how the removal of the lexical constraints on DE works, because replacing DE with other result verbs does not remove the modality involved in the interpretation as the examples above show. Furthermore, if we adopt the step from V-NEG-DE to V-NEG-V in 14th c. CE, it is even more puzzling how our V-DE-V construction emerged in 19th c. CE: the native language learners, centuries after the negative potential form V-NEG-V gained popularity, had to somehow figure out that the modality in the negative potential form was originally introduced by the possibility modal DE and form the ‘positive potential form’ V-DE-V by replacing the negation with possibility modal DE. In Section 5, we discuss an alternative hypothesis where V-DE-V is derived from V-NEG-DE by losing the negation requirement and adding a result verb.

Based on what we have discussed in this section, we can now update our graph of the two-pathway architecture:



Graph 2: The Two-pathway Architecture of the V-DE Construction (Second Pass)

4 Formal Analysis: V-DE-NP/CP/AP

4.1 V-DE-NP/CP: DE in serial verb constructions

We start with the observation that in many cases of the V-DE-NP construction in 9-10th c. CE, DE retains its lexical semantics of transfer of possession. To illustrate with a few examples:

- (28) a. tan de jun ji, su bian zao hui, yu gong zhong shang.
scout DE military intelligence quickly then early return give sir heavy reward
‘(Once you) spy out and obtain military intelligence, return quickly and early, (and I) will give you, sir, a generous reward.’
- b. bian guo qi jiang, qu de xian yao, jin shang huangdi.
then pass PRON river fetch DE celestial medicine enter up emperor
‘He crossed the river, acquired the elixir of immortality, and presented it to the emperor.’
- c. chuchu souxun, ruo ye zhuo de shiseng, su ling jianglai jian wo.
place.DUP search if also seize DE master.monk quickly lead bring.come see I
‘Search everywhere, and if (you) seize and capture the monk, quickly bring him here to see me.’
(*Dunhuang Bianwen Ji*, 9-10th c. CE)

In the first example above, V-DE takes a clear NP object. *Jūnjī* ‘military intelligence’ is a contentful, referential NP, which is exactly the kind of object the lexical verb DE selects. The two verbs form a serial verb construction: *tàn* ‘spy out, scout’ and DE share the implicit subject (the addressee) and share *jūnjī* as the

object of the complex event. The semantic relation is causal in the lexical sense: ‘spy out and (thereby) obtain’. Similarly, in the second example, *qǔ* ‘to fetch’ and DE form a serial verb construction, having the same subject and the same NP object *xiānyào* ‘the elixir of immortality’; DE contributes its transfer-of-possession verb meaning and the whole construction means ‘to fetch and (thereby) obtain’, encoding the same causal relation between the two verbs. The last example can be analyzed in the same manner.

Therefore, in order to characterize the semantics of DE in the V-DE-NP constructions, we need to build on the semantics of the basic transfer-of-possession meaning of DE. To that end, we import the denotation of the lexical verb DE proposed in Bian & Jeretič 2026:

$$(29) \quad \llbracket \text{Ling DE benevolence} \rrbracket^w = \exists e. \exists e'. \text{CAUSE}_w(e', e) \wedge \exists x. \text{benevolence}_w(x) \wedge R_w(\text{Ling}, x, e)$$

The denotation above models the semantics of transfer of possession with Ling as the recipient and benevolence as the theme. In an event of transfer of possession, two implicit events are involved: one causing event that initiates the transfer of possession, and one result event in which the recipient and the theme establish some kind of relation. The relation R_w is designed to be underspecified and is supplied by the context, inheriting from the rich semantic taxonomy of possession relation (see a comprehensive discussion in Myler 2016 Chapter 2 and a summary in Heine 1997:34-35).

We now work on deriving the semantics of DE in V-DE-NP based on (29). Our discussions of the semantic relation of the two verbs in the V-DE-NP constructions exemplified in (28) suggest that the first verb in V-DE-NP constructions serves to identify the causing event that initiates the transfer of possession. For example, ‘to scout’ causes that ‘the addressee enters a relation of abstract possession (to borrow Heine’s 1997 term) with military intelligence’; ‘to fetch’ causes that ‘the subject enters a relation of temporary possession with the elixir of immortality’; etc. Since the serial verb constructions are typically semi-lexicalized, we propose a denotation for V-DE, where V identifies the causing event that initiates the transfer of possession:

$$(30) \quad \llbracket \text{V-DE} \rrbracket^w = \lambda P_{\langle e, t \rangle}. \lambda y. \exists e. \exists e'. V(e') \wedge \text{CAUSE}_w(e', e) \wedge \exists x. P_w(x) \wedge R_w(y, x, e)$$

The underspecified relation R_w captures cases of the V-DE-NP construction in which the meaning of DE is more opaque, i.e., DE does not clearly contribute its meaning of ‘to get; to obtain’. To illustrate with a few examples:

- (31) a. zhi zhi tian ming, zao de yi si.
straight arrive day bright build DE one temple
‘At sunrise, a temple was built.’
b. dan de yi kaolao mantou...
carry.with.a.shoulder.pole DE one basket steamed.bun
‘Carrying a basket of steamed buns on the shoulder with a pole, ...’
c. dao de nan an...
arrive DE south bank
‘Arriving at the south bank, ...’ (Dunhuang Bianwen Ji, circa 9-10th c. CE)

In (31-a), V-DE also takes a clear NP object *yí-sì* ‘one temple’, but DE does not transparently contribute the meaning of ‘to get; to obtain’, because the builders who have built the temple do not own it. Nonetheless, the builders and the temple do enter some kind of relation in the result state, which is identified exactly by the contextually supplied relation R_w . Such relation might be difficult to define exactly, but it can also be found in English (transfer-of-)possession expressions. For example:

- (32) Ieoh Ming Pei designed the building of Four Seasons Hotel New York. He {has got; has} another hotel in Guangzhou, too.⁵

In the English example above, the famous architect Ieoh Ming Pei does not *own* the hotel in Guangzhou; rather, the relation is more aligned to ‘designer-product’, which can be identified by the underspecified relation R_w supplied by the context.

Similarly, in (31-c), v-DE also takes an NP object *nán-àn* ‘the south bank’, but DE clearly does not contribute the meaning of ‘to obtain’. Rather, the interpretation suggests that the subject enters a locative relation with the NP complement. Again, the underspecified R_w in the denotation (30) we proposed can readily identify this kind of relation; English (transfer-of-)possession expressions frequently encode locative relations as well. For example:

- (33) This city {has; has got} some nice nature around it.⁶

In this English sentence, the relation R_w can be easily identified by the interlocutors with the help of the locative PP in the sentence.

We now turn to the v-DE-CP constructions. Recall that this subtype of v-DE construction is first observed along with the v-DE-NP constructions in the same time period, 9-10th c. CE. The interpretation of the v-DE-CP constructions is fairly uniform: the complement CP encodes the actual result/outcome of the verb preceding DE. To illustrate with some examples:

- (34) a. mi de Xiang wang pao guo wei.
bewitch DE Xiang king abandon state position
‘She bewitched King Xiang so that he cast aside his state position.’
b. you xianfeng ma tan de Xiao-Mohe ling jun ershi yu wan.
have vanguard horse(man) scout DE Xiao-Mohe lead troop twenty surplus myriad
‘There were vanguard horsemen (who) spied out (and discovered) that Xiao-Mohe was leading more than two hundred thousand troops.’
c. wu dao jiangjun wen yu, xia de jia bei han liu.
five path general hear word frighten DE soak back sweat flow
‘When the General of the Five Paths heard these words, he was so frightened that that sweat streamed down and his back was soaked.’

(*Dunhuang Bianwen Ji*, circa 9-10th c. CE)

The three examples above are similar to v-DE-NP constructions in terms of the interpretational properties characterized by the denotation in (30) in the following aspects: the verb preceding DE identifies a causing event, and the result event involves a certain relation between the subject and the proposition encoded by the complement clause. For example, in (34-a), the event of bewitching causes the result event in which the subject (the bewitcher) and the complement clause (King Xiang cast aside his state position) stand in some kind of relation identified by the contextually supplied R_w . In (34-b), the event of scouting causes the result event in which the vanguard horseman enters a relation, presumably abstract possession, with the information conveyed by the complement clause ‘Xiao-Mohe was leading more than two hundred thousand troops’. Therefore, we propose a variant of (30) that preserves the semantic mechanisms but differs in the semantic type of the argument it takes:

⁵I am grateful to Don Ringe for this kind of English examples.

⁶I am grateful to Paloma Jeretič for this kind of English examples.

$$(35) \quad \llbracket \text{V-DE}_{\text{CP}} \rrbracket^w = \lambda P_{\langle s, t \rangle} . \lambda y . \exists e . \exists e' . V(e') \wedge \text{CAUSE}_w(e', e) \wedge \exists s . P_w(s) \wedge R_w(y, s, e)$$

(35) minimally varies with (30) in that the CP variant of v-DE takes a proposition of type $\langle s, t \rangle$ as its argument, while the NP variant takes a property of type $\langle e, t \rangle$ as argument. Cross-linguistically, the NP/CP variation under a single verb is a pervasive and well-understood phenomenon typically approached from the perspective of concealed questions (e.g., Heim 1979 and Frana 2017; see also Grimshaw 1979 on verbs selecting semantic types instead of syntactic categories and Chierchia 1984 on type-shifting clausal arguments into entity-correlates of propositions). For example, there is a class of verbs in English that takes both NP arguments and CP arguments (declarative or interrogative): ‘know’, ‘believe’, ‘reveal’, etc. What we have with v-DE allowing both NP and CP complements can be plausibly analyzed as the same phenomenon, especially when we consider the pair (28-a) and (34-b) (repeated below), which have the same verb preceding DE but allow different complement types:⁷

- (36) a. tan de jun ji, su bian zao hui, yu gong zhong shang.
 scout DE military intelligence quickly then early return give sir heavy reward
 ‘(Once you) spy out and obtain military intelligence, return quickly and early, (and I) will give you, sir, a generous reward.’
 b. you xianfeng ma tan de Xiao-Mohe ling jun ershi yu wan.
 have vanguard horse(man) scout DE NAME lead troop twenty surplus myriad
 ‘There were vanguard horsemen (who) spied out (and discovered) that Xiao-Mohe was leading more than two hundred thousand troops.’
 (*Dunhuang Bianwen Ji*, circa 9-10th c. CE)

4.2 v-DE-AP: derivation from DE-CP

Previous studies on the v-DE construction in Modern Mandarin have primarily focused on the v-DE-AP subtype and its connection to the v-DE-CP subtype, building on the implicit assumption that v-DE remains the same across complement categories, and the interpretational differences are derived by distinct syntactic structures of each subtype. Recall that the core empirical puzzle is illustrated by the following sentences:

- (37) a. wo pao de hen kuai.
 I run DE very fast
 ‘I run very fast.’
 b. tamen tiao de hen lei.
 they jump DE very tired
 ‘They jumped till they got very tired.’
 c. tamen ku de shoupa dou shi le.
 they cry DE handkerchief also wet ASP
 ‘They cried so much that even the handkerchief got wet.’ (Huang 1988)

⁷Interestingly, English ‘get’ allows both NP and CP arguments as well, although one might need to further scrutinize whether the following pair of examples involves the same semantic mechanisms as the pair of v-DE-NP and v-DE-CP in Chinese does:

- (i) a. John got the joke.
 b. John got that Mary left.

In the examples above, the (a) variant is called ‘descriptive complement construction’; the (b) and (c) variants are called ‘resultative complement construction’, following the classification tradition in Huang 1988. One crucial structural difference besides the semantic distinction is that resultative complement constructions allow an overt NP following *v-DE*, but descriptive complement constructions do not. To reconceptualize the classification in terms of the complement categories we have been adopting, the (a) and (b) variants are *v-DE-AP* constructions, but the (c) variant is a *v-DE-CP* construction, although the (b) variant patterns more like the (c) variant with respect to the interpretation. C. Li 2015 has proposed a syntactic analysis that captures the structural difference, with the following bracketed structures corresponding to each of the variants above:⁸

(38) Structures of the *V-DE* construction (C. Li 2015:(70))

- a. [_S NP [_{VP} *V-de*] AdvP]]
- b. [_{S1} NP [_{VP} *V-de* [_{S2} Ø PredP]]] (Ø=null element)
- textitc. [_{S1} NP1 [_{VP} *V-de* [_{S2} NP2 PredP]]]

The proposed structures have the following implications regarding what surfaces as *v-DE-AP* constructions, for example the (a) and (b) variants in (37). First, the (a) variant does not allow an overt NP following *v-DE* because the AP is an adjunct to *v-DE* (also argued for in Ernst 2022), while the (b) variant allows a null element following *v-DE* to be the subject of the predicative AP. Second, the AP in the (a) variant modifies the verb; the predicative AP in the (b) variant modifies the null element, typically a null *pro* controlled by the main subject (see e.g., Huang 1984, Huang 1987, and Y.-H. A. Li 2014 for discussions on empty categories in Mandarin). The proposal correctly captures the empirical patterns with diagnostics using *ziji* ‘self’ following *v-DE* and with clausal modifiers like *jintian* ‘today’ as a temporal adverb:

- (39) a. wo pao de (#ziji) hen kuai.
I run DE self very fast
‘I run very fast.’
- b. tamen tiao de (ziji) hen lei.
they jump DE self very tired
‘They jumped till they got very tired.’
- (40) a. wo pao de (#jintian) hen kuai.
I run DE today very fast
‘I run very fast today.’
- b. tamen tiao de (jintian) hen lei.
they jump DE today very tired
‘They jumped till they got very tired today.’

In (39), the (a) variant does not allow *ziji* ‘self’ after *v-DE* but the (b) variant does, indicating that the AP in the (a) variant is an adjunct modifying the verb, while the AP in the (b) variant is predicating a null *pro* controlled by the main subject. Further, the *jintian* ‘today’ diagnostic in (40) suggests that the (a) variant does not have a full CP structure embedded under *v-DE*, while the (b) variant does.

Following this line of reasoning, we build on C. Li 2015 and distinguish between two types of surface

⁸C. Li 2015 labels the AP complement of *v-DE* as adverb phrases. We will continue using the label AP because there is not a clear distinction between nominal modifiers and verbal modifiers in Chinese; Huang et al. 2009 (Chapter 1) does not list adverbs as a distinct category in Chinese.

V-DE-AP constructions: one is underlyingly V-DE-AP, while the other is underlyingly V-DE-CP with a null *pro* as the subject of the embedded CP. Now, we face a puzzle from the perspective of diachronic formal semantics: while the semantics of (38-b) can be derived naturally from our proposal for the V-DE-CP subtype in (35), it is not yet clear how we can interpret the structure in (38-a), and whether there is a derivational relation between (38-a) and (38-b). We address this puzzle in the remainder of this section.

Going back to our historical corpus results, one crucial observation is that the earliest instances of the V-DE-AP construction pattern with the resultative complement construction, where the AP serves as descriptions of the subject, rather than manner or degree of the verb. In 13th c. CE, when V-DE-AP was first observed, the range of available APs include *kexi* ‘delightful’, *keai* ‘lovely’, *hao* ‘good’, and *shifen-beilai* ‘very cheeky’, which are all descriptions of the subjects. The verbs preceding DE are constrained to *sheng* ‘to born’ and *zhang* ‘to grow’. Some data points are replicated below:

- (41) a. ta you yi ge nv'er, jin'nian qi sui, sheng de kexi, zhang de ke'ai.
 he have one CL daughter this.year seven year.old born DE delightful grow DE lovely
 ‘He has a seven-year-old daughter who is delightful and lovely.’
 (Dou'e Yuan, mid-13th c. CE)
- b. bu si zhe xiao nizi sheng de shifen beilai.
 NEG resemble this little girl born DE very cheeky
 ‘They are not like this little girl who is very cheeky in nature.’
 (Dou'e Yuan, mid-13th c. CE)
- c. bi you sheng de hao de nv'ren.
 must have born DE good REL woman
 ‘There must be women who are beautiful.’
 (Bao Daizhi Zhi Zhan Luqilang, mid-13th c. CE)

The interpretation of the examples above patterns exactly with Modern Mandarin sentences having the structure in (38-b): there is a full CP embedded under V-DE with null *pro* subject and adjectival predication, and therefore the AP modifies the subject instead of the verb preceding DE. One hypothesis plausibly follows: with certain verbs like *sheng* ‘to born’ and *zhang* ‘to grow’, V-DE-CP with null subject and adjectival predication became popular, constituting the first instances of what we annotated as V-DE-AP in our corpora. One empirical implication of this hypothesis is that true V-DE-AP constructions with the structure in (38-a) emerged later in time than surface V-DE-AP but underlying V-DE-CP constructions with the structure in (38-b), and it is possible that the former is derived from the latter. Further, if the derivational relation holds, we should expect cases that are ambiguous between the two structures. We do, in fact, observe such cases; for example:

- (42) shi gai zao de hao a!
 this dome build DE good INTJ
 ‘This dome was built so well!’/ ‘They truly built a good dome!’
 (Romance of the West Chamber, 14th c. CE)

The example sentence above is ambiguous between the structures in (38-a) and (38-b) in that the AP *hao* ‘good’ can either be interpreted as an adjunct to V-DE, modifying *zao*, the action of building, or as an adjectival predicate that has a null subject controlled by the main subject *gai* ‘dome’, describing the quality of the dome. This observation can be naturally accounted for by the hypothesis that identical surface form and ambiguity led to structural reanalysis (Eckardt 2006) from [_{S1} NP [_{VP} V-DE [_{S2} Ø AP]]] to [_S

NP [_{VP} V [DE-AP]], a rebracketing process common in diachronic morpho-syntax cross-linguistically. Note that we deviate from (38-a) proposed by C. Li 2015 in that we assume DE-AP forms an adjunct to the verb, rather than having DE as a clitic to the verb while lacking a concrete meaning itself. This process has some non-trivial implications on the formal semantic treatment of DE: we need to spell out the derivational process going from V-DE-CP to V-DE-AP as well as a denotation for a variant of DE that take AP arguments. We start by examining the interpretation of the structure in (38-b), with ‘they jump DE tired’ as example:

$$(43) \quad \llbracket \text{they jump DE } \textit{pro} \textit{ tired} \rrbracket^w = \exists e. \exists e'. \textit{jump}(\textit{they}, e') \wedge \textit{CAUSE}_w(e', e) \wedge \exists s. \textit{tired}_w(\textit{they}, s) \wedge R_w(\textit{they}, s, e)$$

In the denotation above, we have the variant of V-DE that takes CP complements as in (35). The underspecified R_w establishes the relation between the main subject and the AP. Crucially, the subject ‘they’ appears both as the first argument of R_w and as the subject inside the state s characterized by the AP, because the referent of the null *pro* is the same as that of the main subject. This step might be important in motivating a further process of semantic bleaching, because the originally three-place relation R_w is functionally close to a two-place relation between the result event and the state s - the subject becomes recoverable from the state s itself. Further, since our hypothesis has that DE-AP forms an adjunct, we no longer need the verb preceding DE to identify the causing event as in the NP and CP variants of V-DE in (30) and (35); the causation relation is completely bleached as there is no longer a separate result event:

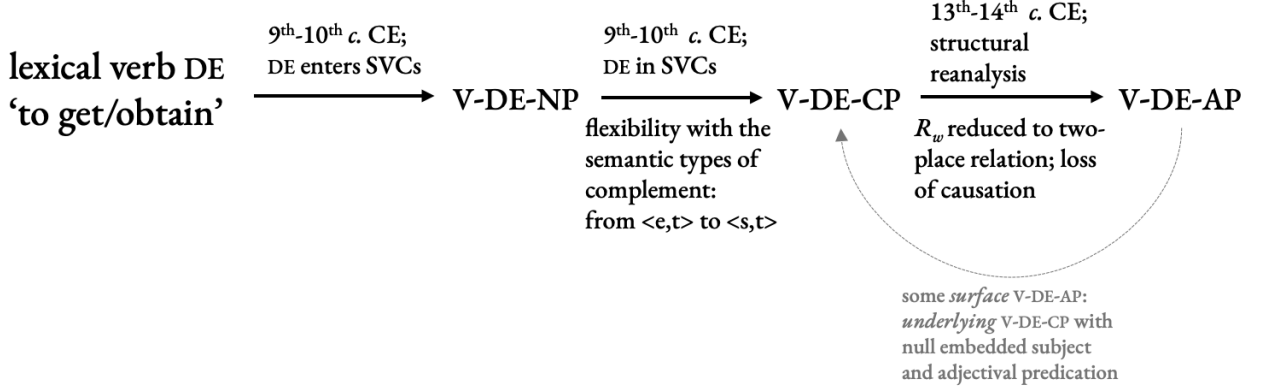
$$(44) \quad \llbracket \text{DE}_{\textit{descriptive}} \rrbracket^w = \lambda g. \lambda e'. \exists s. g(s) \wedge R_w(s, e')$$

In the denotation above, g is the semantics of AP depending on which analysis of adjectives we adopt; e' is the event identified by the verb preceding DE. While the current analysis does not commit us to any specific analysis of adjectives, we can follow Kennedy 2007 by treating adjectives as measure function (on states) and offer a sample derivation of ‘run DE fast’ below:⁹

$$(45) \quad \begin{array}{ll} \text{a.} & \llbracket \textit{fast} \rrbracket^w = \lambda s. \textit{speed}(s); \text{ type } \langle s, d \rangle. \\ \text{b.} & \llbracket \textit{pos} \rrbracket^w = \lambda G_{\langle s, d \rangle}. \lambda s. G(s) \geq d(G). \\ & \text{textit{tc.}} \llbracket \textit{pos fast} \rrbracket^w = \lambda s. \textit{speed}(s) \geq d(\textit{speed}), \text{ where } d(\textit{speed}) \text{ is the contextually de-} \\ & \text{termined standard.} \\ \text{c.} & \llbracket \text{DE POS fast} \rrbracket^w = \lambda e'. \exists s. \textit{speed}(s) \geq d(\textit{speed}) \wedge R_w(s, e'). \\ \text{d.} & \llbracket \textit{run DE POS fast} \rrbracket^w = \lambda e'. \textit{run}(e') \wedge \exists s. \textit{speed}(s) \geq d(\textit{speed}) \wedge R_w(s, e'), \text{ by predicate} \\ & \text{modification.} \end{array}$$

The agent slot can be introduced by Voice or v depending on specific theories and further compose with (45-d). To conclude this section, we offer an updated graph of our two-pathway architecture based on our proposed formal analysis for the V-DE-NP/CP/AP branch (the modal path omitted for clarity):

⁹We import the POS morpheme to introduce degrees because many of the APs allowed in descriptive complementation with V-DE-AP are gradable adjectives. A full analysis that accounts for this observation is left for future research.



Graph 3: The Two-pathway Architecture of the v-DE Construction
(Third Pass; the v-DE-NP/CP/AP Branch)

5 The Modal Pathway and v-DE-v

In this section, I first justify the postulation of a separate modal pathway for the diachronic development of v-DE-v as a subtype of the v-DE construction. Then, I speculate on the diachronic development of v-DE-v by comparing two hypotheses after a brief discussion of the pre-verbal possibility modal DE.

5.1 A separate pathway

The most striking difference between the v-DE-v constructions and the other subtypes of the v-DE construction lies on the interpretational side: v-DE-v constructions always have modalized meaning, while v-DE-NP/CP/AP constructions only have non-modalized readings (cf. Footnote 3 and Lam 2023 for the modal readings of v-DAK-NP in Cantonese). In other words, the argument structure of DE is different for the case of v-DE-v: the argument of DE is not a property or proposition; rather, the preajcent verb phrase is what the modal DE scopes over. This plausibly follows from the assumption that the v-DE-v construction is derived from the possibility modal DE, which was already available in Late Archaic Chinese. In comparison, postulating that the v-DE-v construction is also derived from the lexical verb DE after the v-DE-NP/CP/AP constructions makes the source of modality involved in v-DE-v difficult to account for.

Further, from the perspective of the chronology, we have seen from the historical corpus results in Section 2 that v-DE-NP/CP/AP constructions appeared relatively close in time after serial verb constructions gained popularity after 8th c. CE: v-DE-NP and v-DE-CP are first observed in 9-10th c. CE, and v-DE-AP are observed later in 13-14th c. CE, with ambiguous cases showing continuity between the subtypes. In comparison, the first instances of v-DE-v are observed in 19th c. CE; there is no indication that v-DE-v has any derivational relation to v-DE-NP/CP/AP. If the modal and the non-modal pathways were the same pathway, one would expect more chronological continuity, not the staggered emergence pattern our historical corpora show.

Finally, the distributional properties of DE in v-DE-v differ systematically from those in the v-DE-NP/CP/AP pathway, in ways that are hard to explain if both come from the same source. As we have seen in previous sections, the v-DE-v constructions can be negated by replacing DE with negation; some examples are replicated below:

- (46) wo qu ti shaonainai shefa, zuo de cheng ye bie huanxi, zuo bu
 I go on.behalf.of young.mistress find.a.way do DE complete also NEG.IMP rejoice do NEG
 cheng ye bie manyuan.
 complete also NEG.IMP complain
 ‘I’ll go and try to find a way for the young mistress. If I can finish the job, don’t be overjoyed; if
 I cannot, don’t complain either.’
 (*Journey of Laocan*, late CE 19th c.)

The example above contains a positive V-DE-V construction and its negative counterpart in the same sentence. The negation strategy still holds in Modern Mandarin:

- (47) a. Zhangsan chi de wan zhe wan fan.
 Zhangsan eat DE finish this CL rice
 ‘Zhangsan can eat this whole bowl of rice.’
 b. Zhangsan chi bu wan zhe wan fan.
 Zhangsan eat NEG finish this CL rice
 ‘Zhangsan cannot eat this whole bowl of rice.’

In comparison, the V-DE-NP/CP/AP constructions simply resist negation of any kind.¹⁰ The fact that the modal pattern has a negation strategy that the other patterns lack is strong distributional evidence for separate development trajectories. One further piece of distributional evidence is relevant to the phonological split of DE in Modern Mandarin we discussed in Section 1: some Modern Mandarin speakers allow DE to be pronounced as *dé* with the rising tone in V-DE-V constructions only (coinciding with the phonological realization of the possibility modal DE); speakers generally agree with the judgment that DE in V-DE-CP/AP can only be realized as the neutral tone *de*.¹¹

To sum up, converging interpretational, chronological, and distributional evidence justifies our postulation that the V-DE-V construction was derived from the possibility modal DE. This is a separate pathway from the development of the V-DE-NP/CP/AP constructions, which are derived from the transfer-of-possession verb DE.

5.2 Possibility modal DE and V-DE-V

Before speculating on the development of the V-DE-V construction, recall our discussions of the preverbal possibility modal DE in Late Archaic Chinese (Section 2.2). As we have seen previously, possibility modal uses of DE, along with the lexical verb uses, already became popular in Late Archaic Chinese:

- (48) a. qiu ren er de ren, you he yuan?
 seek benevolence then DE benevolence again what complaint
 ‘Those who seek benevolence and attain it — what complaint could they have?’ (*Lunyu*)
 b. bu de qi jiang, bu shi.
 NEG DE PRON sauce NEG eat
 ‘(Talking about food) If one does not get the proper sauce, they do not eat (the food).’
 (*Lunyu*)

¹⁰With V-DE-AP, one can of course add a negative prefix to the AP to express the negative counterpart; with V-DE-CP, one can negate the embedded CP. But this is not negating the V-DE-AP/CP construction *per se*.

¹¹We do not consider the V-DE-NP construction in Modern Mandarin because it is now obsolete; only a few highly-lexicalized V-DE combinations can take NP complements.

- c. de jian junzi zhe, si ke yi.
DE see gentleman NOM then sufficient P
'To get to meet a true gentleman - that alone is enough.' (Lunyu)
- d. kongzi xia, yu yu zhi yan. qu er bi zhi, bu de yu zhi yan.
Confucius get.off want with PRON talk leave to avoid PRON NEG DE with PRON talk
'Confucius got off the carriage and wanted to talk with Jieyu, but Jieyu went away to avoid Confucius, so Confucius wasn't able to talk to him.' (Lunyu)

According to the corpus results reported in Bian & Jeretič 2026, in Late Archaic Chinese, the flavors of the possibility modal DE were restricted to circumstantial (55.32%), deontic (29.79%), and ability (14.89%); the flavors expanded to teleological and bouletic after 16-17th c. CE.¹²

In Modern Mandarin, the possibility modal use of DE is highly restricted in its distribution. It can only appear in (doubly-)negated environment or co-occur with *zhi* 'only', and the flavor is restricted to circumstantial and deontic. Some examples:

- (49) a. xuesheng bu de chouyan.
student NEG DE smoke
Students are not allowed to smoke. (Deontic possibility)
- b. zhe zhi duiwu wuzi haojin, {zhi de; bu de bu} fanhui.
this CL team supply depleted only DE NEG DE NEG return
'This team's supply was depleted, so they had no choice but to return.'
(Circumstantial possibility)

As we will see, modal flavors make important predictions with respect to the two hypotheses for the development of V-DE-V, which we lay out now. Following our discussion in Section 3, the possibility modal DE entered resultative constructions solely with its negated form in 13th c. CE according to Shi 2023. In this paper, we put aside the puzzle of why the positive form of modal DE could not be used in resultative constructions in the first place.¹³ Rather, we compare two hypotheses on how the V-DE-V was

¹²To follow up with Footnote 2, the possibility modal DE was derived from the lexical verb DE following the type-flexible existential semantics analysis proposed in Jeretič 2021, where the semantics of (transfer-of-)possession and the semantics of modality are closely related:

- (i) $\llbracket \text{Ling DE benevolence} \rrbracket^w = \exists e. \exists e'. \text{CAUSE}_w(e', e) \wedge \exists x. \text{benevolence}_w(x) \wedge R_w(\text{Ling}, x, e)$
- (ii) $\llbracket \text{Ling DE go} \rrbracket^w = \exists e. \exists e'. \text{CAUSE}_w(e', e) \wedge \exists w'. \text{Ling-go}(w') \wedge R_w(w, w', e)$

The semantic mechanisms involved in the denotations above are parallel. The crucial difference is that R_w encodes relation between individuals in the former but relation between worlds in the latter; R_w being underspecified in the latter captures the flexibility of modal flavors expressed by the possibility modal DE.

¹³One possibility is that the primitive form of resultative constructions (separable resultative constructions as in Shi 2023) does not allow monosyllabic phrases to be the result. For example, in the resultative constructions with *cheng* 'to finish' we have seen:

- (i) a. qiu zhi bi zuo shu, shu cheng.
request paper pen write letter letter complete
'(He) requested paper and pens to write a letter and the letter was complete.'
(Shi Shuo Xin Yu, 450 CE)
- b. He Pingshu zhu Laozi shi cheng.
He Pingshu annotate Laozi just complete
'He Pingshu just finished annotating Laozi.'
(Shi Shuo Xin Yu, 450 CE)

derived after 13th c. CE.

The first hypothesis follows from the derivation of the negative potential form V-NEG-V in Shi 2023. The process is roughly as the following: in 13th c. CE, resultative constructions with *bu-de* (NEG-DE) as the result became increasingly popular, as a result, the lexical restriction on DE in V-NEG-DE(-OBJECT) was gradually removed; subsequently, a new grammatical device was established by 14th c. CE - V-NEG-V, dubbed as the negative potential form; finally, in 19th c. CE, the V-DE-V construction emerged as the positive counterpart of the negative potential form. The derivation is represented by the chain below:

(50) Hypothesis 1: V-NEG-DE $\rightarrow$ V-NEG-V $\rightarrow$ V-DE-V

Two theoretical questions immediately arise with this hypothesis as we have discussed in Section 3. Regarding the first step of change, the mechanisms involved in the removal of the lexical constraint on DE is unclear; high frequency alone does not account for the observation that removing the modal DE does not affect the modalized reading of the resulting construction V-NEG-V. Regarding the second step of change, it is not clear how native language learners could figure out that the modality in the negative potential form V-NEG-V was originally introduced by the possibility modal DE, instead of other possibility modals like *kě*, *néng*, etc.

The second hypothesis involves the following process: the resultative V-NEG-DE first lost its requirement for negation and therefore its *bu-de*-resultative reading, yielding V-DE; then the structure can combine with a new result verb, forming V-DE-V. Importantly, the negative potential form does not appear as a step in the development of V-DE-V. The derivation is represented by the chain below:

(51) Hypothesis 2: V-NEG-DE $\rightarrow$ V-DE $\rightarrow$ V-DE-V

In order to argue for Hypothesis 2, one needs to answer the following two theoretical questions: first, what motivates the loss of the requirement for negation? Second, what motivates the addition of a result verb to V-DE?

I will not be able to offer concrete arguments for or against either of the two hypotheses in this paper, but we can find indirect evidence that leans towards Hypothesis 2; the theoretical questions surrounding Hypothesis 2 also appear to be more solvable than those surrounding Hypothesis 1. We now go into further details with the implications of the two hypotheses.

We start with the observation that the available modal flavor in the *v-de-v* construction is restricted to circumstantial in Modern Mandarin, although for the standard, pre-verbal possibility modal DE, both circumstantial and deontic flavors are available.¹⁴

In the (a) sentence, the object of the main verb is duplicated so that the monosyllabic result verb *cheng* has a subject, and the result phrase becomes disyllabic. In the (b) sentence, the result verb *cheng* combines with an adverb to form a disyllabic phrase. A bare, positive possibility modal DE is a monosyllabic word; by being negated, the result phrase becomes disyllabic with *bu-de*. This is purely speculative at this time and certainly needs further empirical evidence to argue for or against.

¹⁴Interestingly, Cantonese *dāk* in V-DAK-V allows both circumstantial and deontic readings:

- (i) keoi5 haang4 dak1 jap6 heoi3.
 3.SG walk DE in go
 ‘S/he is allowed to walk in there.’
 ‘S/he is able to walk in there.’

(Cheng & Sybesma 2004:421 via Lam 2023)

Cantonese *dāk* and Mandarin DE have some intriguing parallelisms and distinctions that are worth investigating in future research.

- (52) ta zou de guoqu nali.
 3.SG walk DE pass-go there
 Attested reading: ‘S/he is able to walk past there.’
 Unattested reading: ‘S/he is allowed to walk past there.’

The sentence above lacks a deontic reading. This observation on modal flavors can potentially tease apart our two hypotheses: on the one hand, Hypothesis 1 predicts that the available range of flavors of DE in the V-DE-V construction would always mirror the modal flavors allowed in the negative potential form V-NEG-V, because under this hypothesis, DE and NEG are structural parallels, forming the positive and negative variants of the same modalized construction. On the other hand, Hypothesis 2 predicts that the available range of flavors of DE in V-DE-V does not correspond to the modal flavors allowed in the negative potential form V-NEG-V because they do not have any derivational relation and developed via independent paths, and therefore modal flavor asymmetries are allowed between V-DE-V and V-NEG-V. Modern Mandarin data patterns are compatible with the predictions of Hypothesis 2:

- (53) ta de che kai de jin Beijing.
 3.SG POSS car drive DE enter Beijing
 Attested reading: ‘Her/His car can be driven to Beijing (from here, because its fuel tank is large enough).’
 Unattested reading (inter-speaker variation): ‘Her/His car is allowed to enter Beijing.’
- (54) Ta de che kai bu jin Beijing.
 3.SG POSS car drive NEG enter Beijing
 Attested reading 1: ‘Her/His car cannot be driven to Beijing (from here, because its fuel tank is too small).’
 Attested reading 2: ‘Her/His car is not allowed to enter Beijing.’

(53) further suggests that with the V-DE-V construction, the deontic reading is unavailable. Its negative counterpart, (54), suggests that the negative potential form V-NEG-V allows both circumstantial and deontic readings. Hypothesis 2 is compatible with the modal flavor asymmetry shown above, but Hypothesis 1 is not.

We now speculate on how one might approach the two theoretical questions surrounding Hypothesis 2. In 14th c. CE, one particular verb, *shǐ* ‘to use’, frequently occurred as the main verb in the V-NEG-DE resultative constructions. At that period of time, only the negative form was observed. For example:

- (55) jiangjiu liuxia zhe, bian shi bu de ye ba.
 put.up.with keep ASP if use NEG DE also P
 ‘I’ll put up with keeping (this tael of silver), even if it cannot be used.’
 (*Lao Qi Da*, circa 14th c. CE)

In 16th c. CE, we observe that *shǐ-de* appears without negation with more generalized meaning ‘can be done; can be accepted’, but it was restricted to rhetorical questions:

- (56) wu gong xiaoshou, zensheng shi de?
 without merit receive how use DE
 ‘How can I receive (this gift) without any merit?!’
 (*Plum in the Golden Vase*, circa 16th c. CE)

In 18th c. CE, *shǐ-de* nearly always means ‘can be done; can be accepted’ and crucially appears in regular declarative sentences freely:

- (57) yao mei you san de huaer, jiushi tou shang dai guo de chaixialai ye shi de.
 if NEG have loose REL flower even head top wear ASP REL take.off also use DE
 ‘If there are no loose flowers, then pearls taken off from those that have been worn on the head
 can also be accepted.’

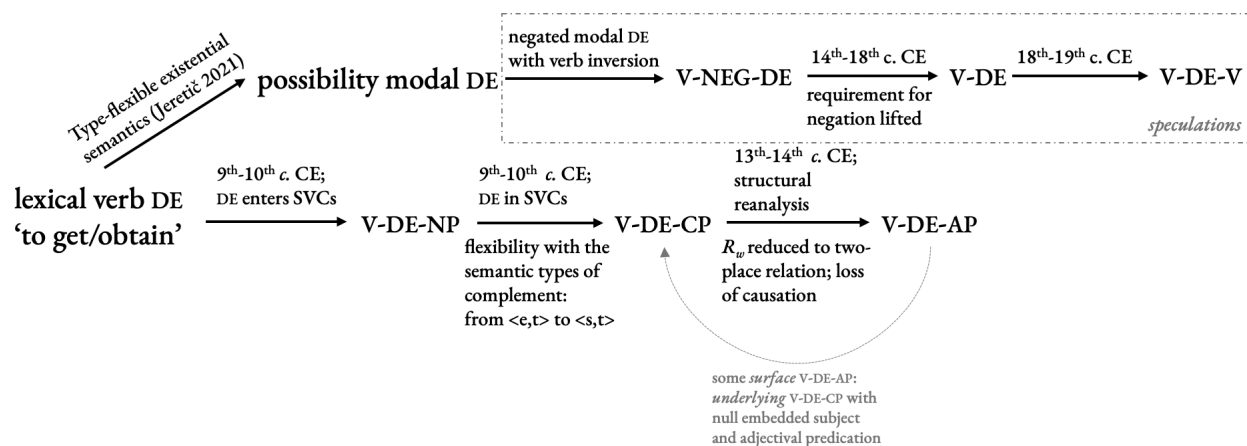
(*Dream of the Red Chamber*, circa 18th c. CE)

The process illustrated above might be able to inform us of how the requirement for negation was gradually lost for the V-NEG-DE resultative construction: the requirement was first weakened to a requirement of environments that implicitly have negative meaning, for example rhetorical questions; and eventually, the requirement of negative environment was also lifted, resulting in the free distribution of the V-DE construction.

If our speculation above is on the right track, then we still have one more question to answer: between 18th c. and 19th c. CE, what motivates the V-DE construction to take an additional result verb?

This question might be difficult to answer if we hold on to the assumption that the V-DE construction is still some kind of resultative construction. However, we can reconceptualize the nature of the original V-NEG-DE resultative construction as a structure of verb inversion triggered by negation, i.e., V-NEG-DE is underlyingly NEG-DE-V, where DE is the standard, pre-verbal possibility modal. Similarly, the positive V-DE construction is underlyingly a standard modal expression DE-V with the verb inverted to precede the modal DE. In this way, the question above is restated as a puzzle of syntactic change, because the V-DE-V construction is underlyingly DE-V-V, with the possibility modal DE taking a serial verb construction as its preajacent, whose semantics can be easily derived building on the semantics of possibility modal DE proposed in Bian & Jeretič 2026 and proposals for the syntax-semantics of serial verb constructions. Whether the syntactic change sketched above can be properly justified is left for future research.

We can now offer our final version of the graph of our two-pathway architecture based on what we have discussed about the modal pathway in this section:



Graph 4: The Two-pathway Architecture of the v-DE Construction (Final)

6 Conclusion

In this study, we investigated the historical development of the V-DE construction in Chinese. The main theoretical claim of the study is the two-pathway architecture where the V-DE-NP/CP/AP subtypes of the V-DE construction are derived from the transfer-of-possession verb DE, while the V-DE-V subtype is derived from the possibility modal DE. The empirical backbone of this claim is converging interpretational, chronological, and distributional evidence drawn from our historical corpora of pre-modern Chinese use of DE: the V-DE-NP/CP/AP subtypes show temporal continuity in their development, do not allow modalized readings, and are incompatible with negation; the V-DE-V subtype emerged many centuries later than the V-DE-NP/CP/AP subtypes, only allows modalized reading, and can be negated with a unique negation strategy. The study also examines the general historical development of verbal phrases in Chinese drawing on Shi 2023 and Shi 2025, arguing that the emergence of serial verb constructions provided the structural basis of the development of the V-DE construction. Based on this structural basis and the semantics of the transfer-of-possession verb DE proposed in Bian & Jeretič 2026, we offered a formal analysis for one of the two branches of our two-pathway architecture, namely the V-DE-NP/CP/AP branch, where we fixed variants of denotation for V-DE and DE with minimal variations motivated by structural changes. Finally, we identified the other branch of the two-pathway architecture, namely the modal branch leading to the emergence of the V-DE-V subtype, as an open question. We speculated on the historical development of the V-DE-V construction by comparing two hypotheses and presented indirect evidence that leans towards the hypothesis where the V-NEG-DE resultative construction first lost its requirement for negation and then acquired a new result verb (V-NEG-DE to V-DE to V-DE-V).

The current study has broader implications on the diachronic formal semantics methodology: how synchronic compositional analysis can be informed by, and constrained by, diachronic evidence. The paper presents a case study where diachronic evidence is a constraint on synchronic analysis, not just a test of it. Previous proposals on the V-DE construction in Modern Mandarin suggest that synchronic analyses of complex constructions are typically underdetermined by synchronic data; multiple compositional analyses can fit the same truth conditions and distributional facts. We show that diachronic evidence can potentially break this underdetermination by illustrating which compositional choices are consequences of independently motivated structural changes.

Our V-DE case study instantiates this methodology. For example, the ambiguity in (42) led to a rebracketing process, which in turn led to the loss of the causal relation and the reduction of R from a three-place relation to a two-place relation. The denotation for DE in (44) is not stipulated to fit the Modern Mandarin facts; it is the minimal modification of (35) consistent with the new structure that emerges from rebracketing. The two-place R, the absence of causation, and the absence of a separate result event are all compositional consequences of the structural change, not merely stipulated to account for the data patterns or to interpret the structure in (38-a). Similarly, the argument type flexibility step from (30) to (35) is another instance of the same methodology: the $\langle e, t \rangle$ to $\langle s, t \rangle$ shift is a minimal variation licensed by the cross-linguistic phenomenon of NP/CP-flexible verbs. Further, the proposed two-pathway architecture itself bears on the same methodology: when synchronic data resists clean unified analysis, the diachronic evidence may reveal that there is no single item or structure to analyze.

There are several potential directions for future research. First, although we argued for the separate modal path in our proposed two-pathway architecture for the V-DE construction based on converging evidence, we still do not have a concrete analysis for the modal path and the development of the V-DE-V construction. The two hypotheses we compared are both established on the assumption that negated

possibility modal *DE*, but not its positive counterpart, entered the *V-NEG-DE* resultative construction. It is not clear why we do not observe bare possibility modal *DE* as the result phrase in resultative constructions. Further, while we presented indirect evidence involving modal flavor asymmetries that leans towards one of the two development hypotheses for *V-DE-V*, the theoretical questions surrounding the more plausible hypothesis are not fully worked out: for one thing, we have empirical evidence for the gradual loosening of the negation requirement, but it is not clear what motivates this kind of structural change; for another, we have restated the step from *V-DE* to *V-DE-V* as a puzzle of syntactic change, but we do not have independent evidence showing that the operation of verb inversion with negation we sketched out is indeed attested in the development of Chinese syntax. Second, the connections between the *V-DE* construction in Mandarin and the *V-DĀK* construction in Cantonese (post-verbal ‘acquire’ constructions in Lam 2023) are particularly interesting. Mandarin *DE* and Cantonese *dāk* are mostly parallel in available functions (e.g., lexical verb, possibility modal, *V-DE-NP/CP/AP/V* constructions) but also show some asymmetries (e.g., *DE* can be used as a necessity modal but not in Cantonese; *dāk* can be used as a focus particle in Cantonese but not in Mandarin). Identifying and analyzing functions of *DE/DĀK* that are parallel but are independently grammaticalized along the same direction (as opposed to shared inheritance from a common ancestor stage or contact-induced borrowing) contributes to the greater cause of characterizing how semantic change works.

References

- Bian, Mingyang & Paloma Jeretič. 2026. *The Story of a Light Verb Modal in Chinese*. NACCL38 at Washington D.C.
- Cheng, Lisa L. -S. & Rint Sybesma. 2004. Postverbal ‘can’ in Cantonese (and Hakka) and Agree. *Lingua*. Variation in the Syntax of Auxiliaries 114(4). 419–445. [https://doi.org/10.1016/S0024-3841\(03\)00067-6](https://doi.org/10.1016/S0024-3841(03)00067-6).
- Chierchia, Gennaro. 1984. *Topics in the Syntax and Semantics of Infinitives and Gerunds*. Amherst, MA: University of Massachusetts Amherst Doctoral Dissertation. <https://www.proquest.com/docview/303307786?sourcetype=Dissertations%20%20Theses>.
- Eckardt, Regine. 2006. *Meaning Change in Grammaticalization: An Enquiry Into Semantic Analysis*. Oxford University Press UK.
- Ernst, Thomas. 2022. “Descriptive complements” are manner adverbials. en. In Andrew Simpson (ed.), *New Explorations in Chinese Theoretical Syntax: Studies in honor of Yen-Hui Audrey Li* (Linguistik Aktuell/Linguistics Today), 111–132. John Benjamins Publishing Company. <https://doi.org/10.1075/1a.272.04ern>.
- Frana, Ilaria. 2017. *Concealed Questions*. Oxford University Press. <https://doi.org/10.1093/acprof:oso/9780199670925.001.0001>.
- Grimshaw, Jane. 1979. Complement Selection and the Lexicon. *Linguistic Inquiry* 10(2). 279–326. <http://www.jstor.org/stable/4178109>.
- Heim, Irene. 1979. Concealed Questions. In Rainer Bäuerle, Urs Egli & Arnim von Stechow (eds.), *Semantics from Different Points of View*, 51–60. Berlin, Heidelberg: Springer. https://doi.org/10.1007/978-3-642-67458-7_5.
- Heine, Bernd. 1997. *Possession: Cognitive Sources, Forces, and Grammaticalization*. eng. Vol. 00083 (Cambridge Studies in Linguistics). Cambridge, U.K.: Cambridge University Press. <https://research.ebsco.com/linkprocessor/plink?id=d47d8209-3ed5-39a4-8cd4-a00bb59cd1ab>.
- Hopper, Paul J. & Elizabeth Closs Traugott. 2003. *Grammaticalization*. 2nd edn. (Cambridge Textbooks in Linguistics). Cambridge: Cambridge University Press. <https://doi.org/10.1017/CB09781139165525>.
- Huang, C.-T. James. 1984. On the Distribution and Reference of Empty Pronouns. *Linguistic Inquiry* 15(4). 531–574. <http://www.jstor.org/stable/4178404>.
- Huang, C.-T. James. 1987. Remarks on Empty Categories in Chinese. *Linguistic Inquiry* 18(2). 321–337. <http://www.jstor.org/stable/4178540>.
- Huang, C.-T. James. 1988. Wǒ pǎo de kuài and Chinese Phrase Structure. *Language* 64(2). 274–311. <https://doi.org/10.2307/415435>.
- Huang, C.-T. James, Y.-H. Audrey Li & Yafei Li. 2009. *The Syntax of Chinese* (Cambridge Syntax Guides). Cambridge: Cambridge University Press. <https://doi.org/10.1017/CB09781139166935>.
- Jeretič, Paloma. 2021. *Neg-Raising Modals and Scaleless Implicatures*. New York: New York University Doctoral Dissertation. <https://ling.auf.net/lingbuzz/006390>.
- Kennedy, Christopher. 2007. Vagueness and Grammar: The Semantics of Relative and Absolute Gradable Adjectives. *Linguistics and Philosophy* 30(1). 1–45. <http://www.jstor.org/stable/29737189>.

- Lam, Cherry Chit-yu. 2023. Rethinking postverbal ‘acquire’ and related constructions in Cantonese: Polyfunctionality and parameters. *Language and Linguistics* 24(4). 674–732. <https://doi.org/10.1075/lali.00143.chi>.
- Li, Chao. 2015. On the V-DE Construction in Mandarin Chinese. en. *Lingua Sinica* 1(1). 6. <https://doi.org/10.1186/s40655-015-0006-4>.
- Li, Chris Wen-Chao. 2016. Classical Chinese. In Sin-Wai Chan & James Minett (eds.), *Routledge Encyclopedia of the Chinese Language*. London: Routledge.
- Li, Wenshan. 2024. The syntactic variety and semantic unity of the V de resultative construction in Mandarin Chinese. *Poznan Studies in Contemporary Linguistics* 60(2). 189–226. <https://doi.org/10.1515/psic1-2023-0042>.
- Li, Yen-Hui Audrey. 1990. *Order and Constituency in Mandarin Chinese*. Joan Maling & Luigi Rizzi (eds.). Vol. 19 (Studies in Natural Language and Linguistic Theory). Dordrecht: Springer Netherlands. <https://doi.org/10.1007/978-94-009-1898-6>.
- Li, Yen-Hui Audrey. 2014. Born empty. *Lingua*. Structural Approaches to Ellipsis 151. 43–68. <https://doi.org/10.1016/j.lingua.2013.10.013>.
- Meisterernst, Barbara. 2019. A semantic analysis of modal DE in pre-modern Chinese. *Lingua Sinica* 5(1). 1–29. <https://doi.org/10.2478/linguasinica-2020-0001>.
- Myler, Neil. 2016. *Building and Interpreting Possession Sentences*. eng. Cambridge, MA: The MIT Press. <https://research.ebsco.com/linkprocessor/plink?id=9b27e515-52a8-34c5-9180-27e51b45ee93>.
- Shi, Yuzhi. 2023. *The Evolution of Chinese Grammar*. Cambridge: Cambridge University Press. <https://doi.org/10.1017/9781108921831>.
- Shi, Yuzhi. 2025. *The Historical Morpho-Syntax of Chinese*. Springer & Zhejiang University Press.
- Tay, Wenkai. 2024. *Resultative expressions in Mandarin Chinese*. London, UK: University College London Doctoral Dissertation.
- Xie, Zhiguo. 2012. The modal uses of de and temporal shifting in Mandarin Chinese. en. *Journal of East Asian Linguistics* 21(4). 387–420. <https://doi.org/10.1007/s10831-012-9093-8>.